

Fan Xiao

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Education

Ludwig-Maximilians-Universität München, Munich, Germany

2023.09 – Now

PhD candidate in Medical Physics

Research focus: AI-accelerated Dose Calculation, Photon/Proton Monte Carlo Simulation, Image Segmentation/Registration

Southern Medical University, Guangzhou, China

2020.09 – 2023.06

MS in Biomedical Engineering, GPA 3.6/4.0

Wuhan University of Technology, Wuhan, China

2016.09 – 2020.06

BS in Communication Engineering

Research Experience

Deep learning (DL)-based dose calculation on MRI versus synthetic CT | Co-first Author 2025.11–Now

- Developed a ConvLSTM based photon segment dose calculation engine on prostate MRI
- Supported a comparative study benchmarking DL-based MRI dose calculation against Monte Carlo (MC) simulations on synthetic CT
- Achieved full VMAT plan dose calculation on MRI in ~5 s with 1%/1mm gamma passing rate > 90%
- To be submitted in *Physics and Imaging in Radiation Oncology* (2026)

DoseRAD2026 MICCAI Challenge Dataset | First Author

2025.09–Now

- Generated and co-organize a public paired CT and MRI dataset (abdomen & thorax patients) with beam-level photon and proton MC dose distributions for developing and evaluating advanced dose calculation methods
- Pre-processing: deformable image registration, air-cavity correction, matRad photon/proton planning
- Ground-truth photon (6 MV) and proton dose distributions were computed using open-source Geant4 MC simulations, yielding 40,500 photon beams and 81,000 proton beamlets
- Submitted in *Medical Physics* (2026), DOI: <https://arxiv.org/abs/2604.12778>

Multi-model study of fast VMAT segment dose calculation | First Author

2025.07–2026.04

- Designed a comparative VMAT dose calculation study evaluating Beam's Eye View (BEV) and patient coordinate methods across representative models (Transformer, CNN-Mamba, CNN-ConvLSTM, C3D and DeepDose-C3D) on prostate patients
- BEV-based methods show more robust segment dose prediction than patient coordinate methods
- Developed two lightweight DL models (CNN-Mamba, CNN-ConvLSTM) achieving segment dose inference time <40 ms with low memory footprint (2.9 GB)
- Published in *Physics in Medicine & Biology* (2026), DOI: [10.1088/1361-6560/ae6413](https://doi.org/10.1088/1361-6560/ae6413)

DL-based synthetic-CT-free photon dose calculation: proof-of-concept | First Author 2024.10 – 2025.06

- Proposed a UNet-LSTM framework to predict photon beamlet dose directly from prostate MRI data
- Explored the domain shift challenge (lack of electron density information in MRI) by integrating CT-MRI deformable registration with air cavity segmentation
- Achieved 12 ms per photon beamlet inference time with a gamma passing rate (2%/2mm) exceeding

99.50% compared to MC dose

- Published in *Medical Physics* (2025), DOI: [10.1002/mp.70106](https://doi.org/10.1002/mp.70106)

Proton Range Verification via LSTM-based Prompt Gamma Prediction | First Author 2024.01 – 2024.09

- Developed an LSTM model by integrating relative stopping power maps and proton dose to predict prompt gamma emissions across wide proton energies (125–210 MeV)
- Achieved sub-millimeter accuracy with a mean absolute range shift deviation of 0.15 mm against MC prompt gamma emissions
- Reduced prediction time to <130 ms per pencil beam, facilitating sub-second feedback for adaptive proton therapy
- Published in *Physics in Medicine & Biology* (2024), DOI: [10.1088/1361-6560/ad8e2a](https://doi.org/10.1088/1361-6560/ad8e2a)

Simultaneous dose and fluence prediction for head and neck IMRT | Co-Author 2022.07 – 2023.05

- Developed a shared-encoder 3D U-Net for simultaneous dose & fluence prediction in H&N IMRT
- Achieved fluence MAE $0.53\% \pm 0.13\%$, SSIM 0.96 ± 0.02 , dose error <1 Gy for target/OAR structures
- Enabled automated IMRT planning pipeline by using predicted dose as optimization objective and fluence as warm-start initialization
- Published in *Radiation Oncology* (2023), DOI: [10.1186/s13014-023-02287-4](https://doi.org/10.1186/s13014-023-02287-4)

Technical Skills

- Programming languages: Python, C++ (Geant4), Bash
- Deep Learning: Pytorch, LSTM, Transformer, Mamba, Unet/nnUnet
- Tools: Git, Docker, Slurm, Cupy, Plastimatch, Pydicom, matRad
- TPS planning experience: RayStation (scriptable planning), Varian Eclipse

Additional Info and Awards

- Conference experience:
ESTRO 2026 Poster (Stockholm), AAPM 2025 Oral (Washington, DC),
ICCR 2024 Oral (Lyon), ECMP 2024 Oral (Munich)
 - Awards:
Chinese National Scholarship (2022) – Top 2%
First-class Academic Scholarship (2022, 2023) – Top 20%
 - Languages:
Chinese (Native), English (Professional Working Proficiency)
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